

Lagrange Interpolating Polynomials

Formula: $\rightarrow L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$

Annotations:
No mentioning x_0 (pointing to numerator)
just here (pointing to denominator)

$\rightarrow L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$ *Should match.*

Polynomial $\rightarrow p(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$

for final Absolute Error (Error Bound):

- ① Approximate E $\Rightarrow$ by substituting $p(x)$ with x that given in Q
- ② Exact E $\Rightarrow$ by substituting $f(x)$ with x that given in Q
- ③ final Absolute Error $\Rightarrow | \text{Appr. E} - \text{Exact E} |$

Q1% 1. For the given functions $f(x)$, let $x_0 = 0$, $x_1 = 0.6$, and $x_2 = 0.9$. Construct interpolation polynomials of degree at most one and at most two to approximate $f(0.45)$ and find the absolute error.

$\hookrightarrow x$

a. $f(x) = \cos x$

3 points $\Rightarrow n=2$

Degree 1% $x_0 = 0$, $x_1 = 0.6$, $x_2 = 0.9$

① $f(x_0) = \cos(0) = 1$
 $f(x_1) = 0.8253356$

② $L_0 = \frac{x - x_1}{x_0 - x_1} = \frac{x - 0.6}{0 - 0.6} = \frac{x - 0.6}{-0.6}$

$L_1 = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{0.6 - 0} = \frac{x}{0.6}$

③ $p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{x - 0.6}{-0.6} (1) + \frac{x}{0.6} (0.8253356)$

Absolute Error % $x = 0.45$

Appr. E $\Rightarrow p(0.45) = \frac{0.45 - 0.6}{-0.6} (1) + \frac{0.45}{0.6} (0.8253356) = 0.8690017$

Exact. E $\Rightarrow f(0.45) = \cos(0.45) = 0.9004471024$

Absolut. E $\Rightarrow | \text{Appr. E} - \text{Exact. E} | = | 0.8690017 - 0.9004471 | = 0.0314454$

Degree 2% $x_0 = 0$, $x_1 = 0.6$, $x_2 = 0.9$

① $f(x_0) = \cos(0) = 1$
 $f(x_1) = 0.8253356$
 $f(x_2) = 0.6216099$

② $L_0 = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 0.6)(x - 0.9)}{(0 - 0.6)(0 - 0.9)} = \frac{(x - 0.6)(x - 0.9)}{0.54}$

$L_1 = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 0)(x - 0.9)}{(0.6 - 0)(0.6 - 0.9)} = \frac{x(x - 0.9)}{-0.18}$

$$L_2 = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x - 0)(x - 0.6)}{(0.9 - 0)(0.9 - 0.6)} = \frac{x(x - 0.6)}{0.27}$$

$$\textcircled{3} \quad p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) \\ = \frac{(x - 0.6)(x - 0.9)}{0.54}(1) + \frac{x(x - 0.9)}{-0.18}(0.8253356) + \frac{x(x - 0.6)}{0.27}(0.6216099)$$

Absolute Error % $x = 0.45$

$$\text{Appr. E} \Rightarrow p(0.45) = \frac{(0.45 - 0.6)(0.45 - 0.9)}{0.54}(1) + \frac{0.45(0.45 - 0.9)}{-0.18}(0.8253356) + \frac{0.45(0.45 - 0.6)}{0.27}(0.6216099)$$

$$p(0.45) = 0.898100075$$

$$\text{Exact E} \Rightarrow f(0.45) = \cos(0.45) = 0.9004471024$$

$$\text{Absolute E} \Rightarrow |\text{Appr. E} - \text{Exact E}| = |0.898100075 - 0.9004471| = 0.0023$$

So degree 2 has better approximation value than degree 1
as well absolute error for degree 2 is less than degree 1

$$\text{b. } f(x) = \sqrt{1+x} \quad x_0 = 0, x_1 = 0.6, x_2 = 0.9$$

Degree 1

$$\textcircled{1} \quad f(x_0) = \sqrt{1+0} = 1$$

$$f(x_1) = \sqrt{1+0.6} = 1.2649110$$

$$\textcircled{2} \quad L_0 = \frac{x - x_1}{x_0 - x_1} = \frac{x - 0.6}{0 - 0.6}$$

$$L_1 = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{0.6 - 0}$$

$$\textcircled{3} \quad p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{x - 0.6}{0 - 0.6}(1) + \frac{x - 0}{0.6 - 0}(1.2649110)$$

Absolute Error % $x = 0.45$

$$\Rightarrow p(0.45) = \frac{0.45 - 0.6}{0 - 0.6}(1) + \frac{0.45 - 0}{0.6 - 0}(1.2649110) = 1.1986832$$

$$\Rightarrow f(0.45) = \sqrt{1+0.45} = 1.2041594$$

$$\Rightarrow \text{absolute } E = |1.1986832 - 1.2041594| = 0.00547$$

Degree 2 % $x_0 = 0, x_1 = 0.6, x_2 = 0.9$

$$\begin{aligned} \textcircled{1} \quad f(x_0) &= \sqrt{1+0} = 1 \\ f(x_1) &= \sqrt{1+0.6} = 1.2649110 \\ f(x_2) &= \sqrt{1+0.9} = 1.3784048 \end{aligned}$$

$$\textcircled{2} \quad L_0 = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-0.6)(x-0.9)}{(0-0.6)(0-0.9)} = \frac{(x-0.6)(x-0.9)}{0.54}$$

$$L_1 = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-0)(x-0.9)}{(0.6-0)(0.6-0.9)} = \frac{x(x-0.9)}{-0.18}$$

$$L_2 = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-0)(x-0.6)}{(0.9-0)(0.9-0.6)} = \frac{x(x-0.6)}{0.27}$$

$$\textcircled{3} \quad p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$$

$$= \frac{(x-0.6)(x-0.9)}{0.54} (1) + \frac{x(x-0.9)}{-0.18} (1.2649110) + \frac{x(x-0.6)}{0.27} (1.3784048)$$

Absolute Error % $x = 0.45$

$$\begin{aligned} \Rightarrow p(0.45) &= \frac{(0.45-0.6)(0.45-0.9)}{0.54} + \frac{0.45(0.45-0.9)}{-0.18} (1.2649110) + \frac{0.45(0.45-0.6)}{0.27} (1.3784048) \\ &= 1.203423728 \end{aligned}$$

$$\Rightarrow f(0.45) = \sqrt{1+0.45} = 1.2041594$$

$$\Rightarrow \text{absolute } E = |1.203423728 - 1.2041594| = 0.0007$$

So degree 2 is better than degree 1

Q58

5. Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$

$\Rightarrow$ find Lagrange and polynomials

$n=3 \Rightarrow 4$ points

degree 1: $x_0 = 8.1$, $x_1 = 8.3$, $x_2 = 8.6$, $x_3 = 8.7$

$$L_0 = \frac{(x - x_1)}{(x_0 - x_1)} = \frac{(x - 8.3)}{(8.1 - 8.3)} = \frac{(x - 8.3)}{-0.2}$$

$$L_1 = \frac{(x - x_0)}{(x_1 - x_0)} = \frac{(x - 8.1)}{(8.3 - 8.1)} = \frac{(x - 8.1)}{0.2}$$

$$p_1(x) = \frac{(x - 8.3)}{-0.2} (16.94410) + \frac{(x - 8.1)}{0.2} (17.56492)$$

degree 2: $x_0 = 8.1$, $x_1 = 8.3$, $x_2 = 8.6$, $x_3 = 8.7$

$$L_0 = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 8.3)(x - 8.6)}{(8.1 - 8.3)(8.1 - 8.6)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 8.1)(x - 8.6)}{(8.3 - 8.1)(8.3 - 8.6)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x - 8.1)(x - 8.3)}{(8.6 - 8.1)(8.6 - 8.3)}$$

$$p_2(x) = \frac{(x - 8.3)(x - 8.6)}{(8.1 - 8.3)(8.1 - 8.6)} (16.94410) + \frac{(x - 8.1)(x - 8.6)}{(8.3 - 8.1)(8.3 - 8.6)} (17.56492) + \frac{(x - 8.1)(x - 8.3)}{(8.6 - 8.1)(8.6 - 8.3)} (18.50515)$$

degree 3: $x_0 = 8.1$, $x_1 = 8.3$, $x_2 = 8.6$, $x_3 = 8.7$

$$L_0 = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} = \frac{(x - 8.3)(x - 8.6)(x - 8.7)}{(8.1 - 8.3)(8.1 - 8.6)(8.1 - 8.7)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} = \frac{(x - 8.1)(x - 8.6)(x - 8.7)}{(8.3 - 8.1)(8.3 - 8.6)(8.3 - 8.7)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} = \frac{(x - 8.1)(x - 8.3)(x - 8.7)}{(8.6 - 8.1)(8.6 - 8.3)(8.6 - 8.7)}$$

$$L_3 = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-8.1)(x-8.3)(x-8.6)}{(8.7-8.1)(8.7-8.3)(8.7-8.6)}$$

$$\begin{aligned} p(x) = & \frac{(x-8.3)(x-8.6)(x-8.7)}{(8.1-8.3)(8.1-8.6)(8.1-8.7)} (16.94410) + \\ & \frac{(x-8.1)(x-8.6)(x-8.7)}{(8.3-8.1)(8.3-8.6)(8.3-8.7)} (17.56492) + \\ & \frac{(x-8.1)(x-8.3)(x-8.7)}{(8.6-8.1)(8.6-8.3)(8.6-8.7)} (18.50515) + \\ & \frac{(x-8.1)(x-8.3)(x-8.6)}{(8.7-8.1)(8.7-8.3)(8.7-8.6)} (18.82091) \end{aligned}$$

Q7%

7. The data for Exercise 5 were generated using the following functions. Use the error formula to find a bound for the error and compare the bound to the actual error for the cases $n = 1$ and $n = 2$.

a. $f(x) = x \ln x$

from Q5 $\Rightarrow$ a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$

$$\begin{aligned} P_1(x) &= \frac{(x-8.3)}{-0.2} (16.94410) + \frac{(x-8.1)}{0.2} (17.56492) \\ &= \frac{(8.4-8.3)}{-0.2} (16.94410) + \frac{(8.4-8.1)}{0.2} (17.56492) = 17.87533 \end{aligned}$$

$$\begin{aligned} P_2(x) &= \frac{(8.4-8.1)(8.4-8.6)}{(8.1-8.3)(8.1-8.6)} (16.94410) + \frac{(8.4-8.1)(8.4-8.6)}{(8.3-8.1)(8.3-8.6)} (17.56492) + \frac{(8.4-8.1)(8.4-8.3)}{(8.6-8.1)(8.6-8.3)} (18.50515) \\ &= 17.87713 \end{aligned}$$

$$f_{\text{act.}} = 8.4 \ln(8.4) = 17.87714633$$

$$P_1(x) = |17.87533 - 17.87714633| = 0.001$$

$$P_2(x) = |17.87713 - 17.87714633| = 0.0001$$